OSTEOARTHRITIS INITIATIVE / SECONDARY ANALYSIS

Structure-Pain Discordance and 48-Month Knee Pain Prediction Error

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MAIN FINDING

The association did not persist after baseline-pain adjustment.

The adjusted result changes the interpretation of the frozen unadjusted association.

01 Question and design

Does disagreement between knee X-ray severity and reported pain identify harder-to-predict knees?

We examined absolute prediction error for 48-month WOMAC pain using the Osteoarthritis Initiative.

8,920

baseline knees from 4,490 participants

8,232

knees with outcomes from 4,146 participants

Analysis

Estimate expected baseline pain within training folds; define discordance as the absolute gap from observed pain.

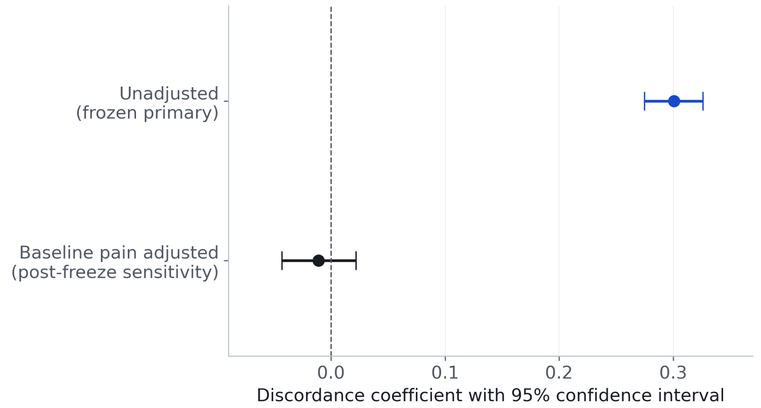
Fit elastic net with participant-grouped nested five-fold cross-validation. Keep both knees of each person together.

Regress held-out absolute errors on discordance with participant-clustered inference.

Baseline replacements excluded. Outcomes were not imputed. Follow-up was missing for 7.7% of baseline knees.

02 Association and adjustment

A positive unadjusted result was not independent of baseline pain.



Frozen primary analysis

Coefficient 0.301; 95% CI 0.275 to 0.326.

Post-freeze sensitivity analysis

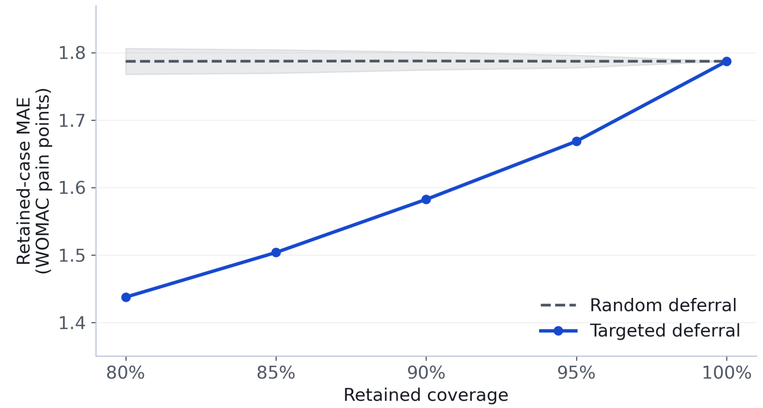
Adjusted coefficient -0.011; 95% CI -0.043 to 0.022; p = 0.52.

Discordance correlated with baseline pain (r = 0.751). Baseline symptom burden is the stronger candidate explanation in these data.

The adjusted analysis was reported in the manuscript after the freeze. It is a post hoc sensitivity, not a prespecified replacement for the primary analysis.

03 Selective prediction

Deferral reduces retained error by setting aside difficult cases.



At 10% deferral

MAE fell from 1.787 to 1.582 among 7,409 retained knees: an 11.47% relative reduction within that subset.

The error model included discordance. This result does not independently confirm a discordance effect or show improved predictions for every knee.

Limits and next step

One cohort; internal validation only. Quartile error means were not monotonic. Subgroup deferral rates varied. External validation is needed; causation and clinical benefit are not established.

Manuscript + code + interactive companion

ksricharan7.github.io/oai-model-reliability/

Sources: author-shared manuscript and frozen aggregate analysis outputs (September 13, 2026). Adjustment is manuscript-reported, post-freeze. Poster redesigned October 15, 2026. Research use only; no acceptance or completed presentation claimed.